

CT-SCAN DATA VISUALIZATION & ML PREDICTION DASHBOARD



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PROJECT OBJECTIVE:

Streamline and automate CT scan analysis through advanced EDA, ML &LLM.

THE NEED FOR AUTOMATED CT-SCAN ANALYSIS

Current situation

Crafting graphs in Excel is tedious and time-intensive.

Manual analysis is error-prone and inefficient.

Need

Simplify and speed up CT scan analysis.

Leverage ML and AI to enhance decision-making.

Challenges

Raw data needs extensive pre-processing to ensure optimal EDA

- slab;tissue[##,##];tissue_stat
- tissue[##,##]-U-tissue[##,##]

METHODOLOGY

INPUT & PROCESS

Upload raw csv file.
Data cleaning, standardization,
and validation

EDA

Interactive
visualization for
easily interpretable
insights

ML

Identify patterns
&
Predict outcomes.

LLM

AI- powered instant
query-based insights

*EDA- Exploratory Data Analysis

*ML- Machine Learning

*LLM- Large Learning Model

ARCHITECTURE AND TECH STACK

Layer	Component	Technology/Tool	Purpose
Frontend	Dashboard Framework	Dash	Create an interactive user interface for data visualization and queries.
	UI Rendering	Plotly	Generate scatter plots, line charts, and histograms.
	Input Handling	Dash Core Components	Enable user inputs (text boxes, dropdowns, buttons).
Backend	Web Framework	FastAPI	Serve RESTful API for LLM interaction and data handling.
	Server	Uvicorn	ASGI server for handling requests efficiently.
	Caching	Flask-Caching	Cache precomputed or frequent results for faster access.
LLM	Pretrained Model	EleutherAI GPT-Neo (1.3B)	Power natural language queries and answers.
	Model Hosting	Hugging Face Transformers	Load and manage pretrained models.
	Hardware Acceleration	Metal Performance Shaders (MPS)	Leverage GPU acceleration for faster inference on macOS.
Data Handling	Data Processing Library	Pandas	Process, clean, and organize CT scan data.
Machine Learning (ML)	Clustering and Prediction	Scikit-learn	Perform unsupervised learning and supervised learning tasks.
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Monitoring	Logging	Python Logging Module	Track and debug application events.
	Profiling	PyTorch Profiler	Identify bottlenecks in AI inference processes.

DASHBOARD DESIGN AND OVERVIEW

CT Scan HU Analysis

Machine Learning

LLM Assistant

CT Scan HU Value Analysis Dashboard

Drag and Drop or Select Files

Select Tissue:

Select...

Select Slab:

Select...

Select Measurement Metric:

Select...

HU Value Threshold Range:

-32768

-3000

-1000

0

1000

3000

Max

Select Patient:

Select a patient ID

Select HU Metric:

☒ HU Mean

☐ HU Max

☐ HU Std

☐ HU Median

☐ HU Min

Data Preview

HU distribution and Time-Series Analysis

Z-Score

Clustering & Segmentation

Vertebrae Count

Tissue Count Plot

Slices thickness and Gender

Uploaded Data Preview

Clear Cache

DATA PREPROCESSING

Segmentation:

- Columns were split using the ‘;’ delimiter to separate key components in sheet. Used it to create the segments based on tissues, slabs, and measurement metrics.
- Dashboard has separate dropdowns to select each segment.

Data Accuracy:

NA values were removed for analysis and prediction to improve accuracy.

Before uploading csv

Select Tissue:

Select...

Select Slab:

Select...

Select Measurement Metric:

Select...

HU Value Threshold Range:

-1258

-1000

-500

0

500

1000

Max

Select Patient:

Select a patient ID

Select HU Metric:

☒ HU Mean

☐ HU Max

☐ HU Std

☐ HU Median

☐ HU Min

Automatic segmentation

Select Tissue:

IMAT	LPMFHEAD	RULWGMAT	THAT	LULWGMAT	TBICLR	RPFMRSHAF	VAT	LLUNG	PANC	RKID	RMASSETER	ALLIMAT	LABONE
ASCITIS	ALLSKM_NORMAS	LTPMFORALS	RAIMAT	SAT	Prostate	RHPHIONE	GB	LULPGMAT	RABONE	LASAT	CAC	RILPOSOAS	RPECMUR
RPECMNR	LASKM	LHPHONE	SPL	LMASSERT	BRAIN	LPMFSHAPT	SKM	RPMRHEAD	Bone	LUNGFUSION	LILPOSOAS	LAIMAT	RULPGSM
BLADDER	AoA	LULPGMSK	PaAt	TRACH	LPMRNECK	RTMPORALIS	HRT	LPECMNr	RASKM	LPECMR	ALLSKM	RASAT	RUPMGIMAT
RLUNG	EpoT	LIV	AOC	LKID	RPMRNECK	LLUWGSKM	RULWGSKM						

Select Shape:

Headstair-Headst	C1stat-C1end	T1stat-T1end	Scannstar-Scannet	C2stat-C2end	T2stat-T2end	T2stat-T2end	T2stat-T2end	L1mid	C2stat-C2end	C2stat-C2end
L1mid-L1mid	C5stat-C5end	FULL_SCAN	T4stat-T4end	T2stat-T2end	T2stat-T2end	L1mid-vrg12	C2stat-C2end	C2stat-C2end	T11stat-T11end	T1stat-T1end
T10stat-T10end	T12stat-T12end	T2stat-T2end	T4stat-T4end	T4stat-T4end	T4stat-T4end					

Select Measurement Metric:

scan_folder	img_size_WxHxI	StudyDate	HU_mean	segmented_tissues	PatientPosition	HU_std	ManufacturerModelName	Manufacturer	cross_sectional_area_pixels
cross_sectional_area_cm2	height_mm	ImageType	verts_in_scan	SeriesDate	vessel_size_WxHxI	Modality	SoftwareVersions	AcquisitionDate	num_slices
BodyPartExamined	verts_with_msd_anno_in_scan	KVP	orig_puh	volume_vessels	HU_max	most_inferior_vert_in_scan	PatientID	volume_cm3	most_superior_vert_in_scan
SeriesNumber	PatientSex	HU_median							

HU Value Threshold Range:

-1776 -3000 -1000 0 1000 3000 Max

Select Patient:

Select a patient ID

Select HU Metric:

● HU Mean ● HU Max ○ HU Std ○ HU Median ○ HU Min

TABS AND FUNCTIONALITY

CT Scan Analysis

Interactive Insights:

HU distribution, Z-Score, and segmentation, tissue counts, vertebrae, and slice thickness.

Machine Learning

Insights through ML:

Feature selection, error analysis, predictions, and correlations.

LLM Assistant

AI powered responses:

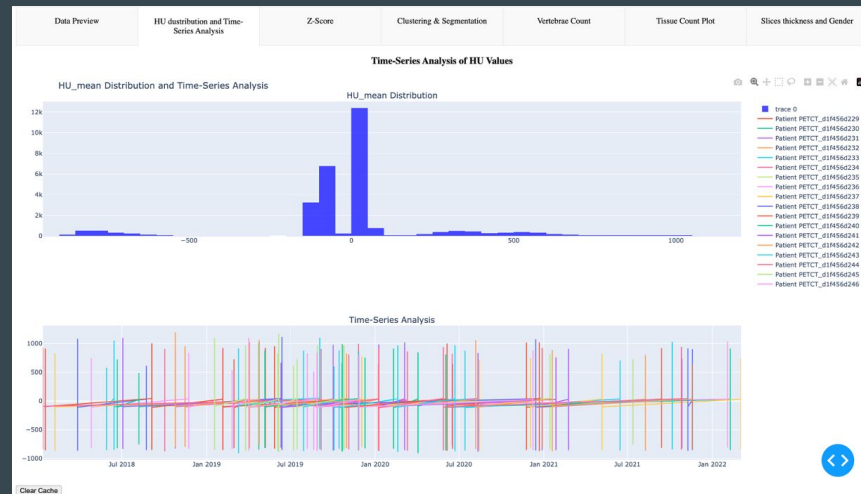
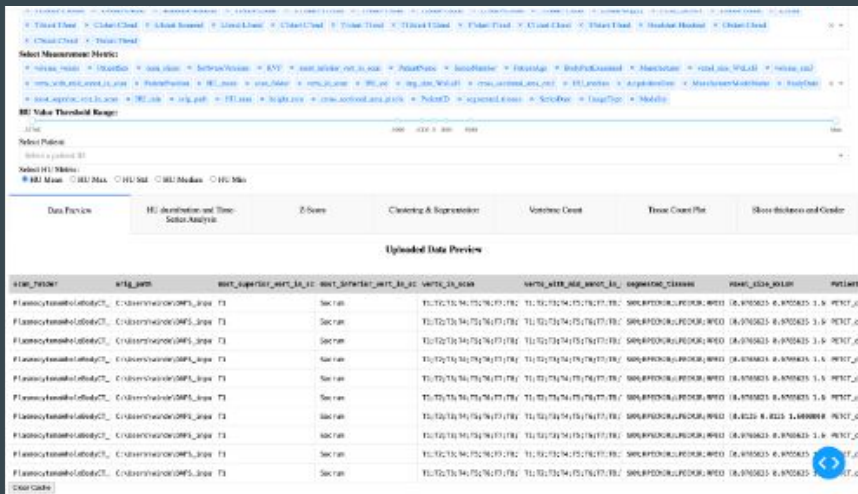
Interactive prompt for CT scan queries.

CT SCAN ANALYSIS AND SUB TABS

- **Data Preview**

View and explore the uploaded CT scan dataset, ensuring it is ready for analysis.

- **HU Distribution** : Analyze Hounsfield Units to uncover patterns
- **Time Series Analysis** : Track HU changes and trends over time or patients



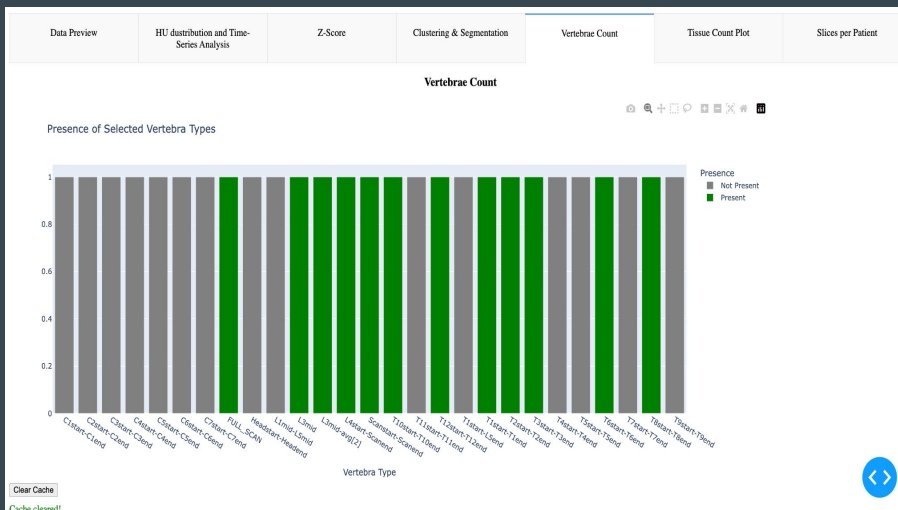
- **Z-Score**

Detect and highlight statistical outliers or deviations in the dataset using Z-scores.



- **Clustering & segmentation**

Automatically group and segment patients according to HU values across scans for easier interpretation.

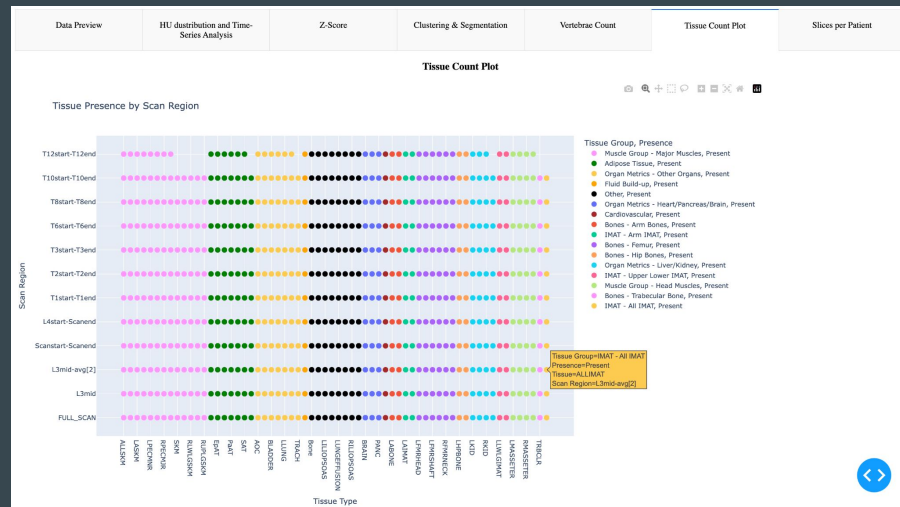


- **Vertebrae Count**

Count the present types of vertebrae scan categories from the scan for specific medical analysis

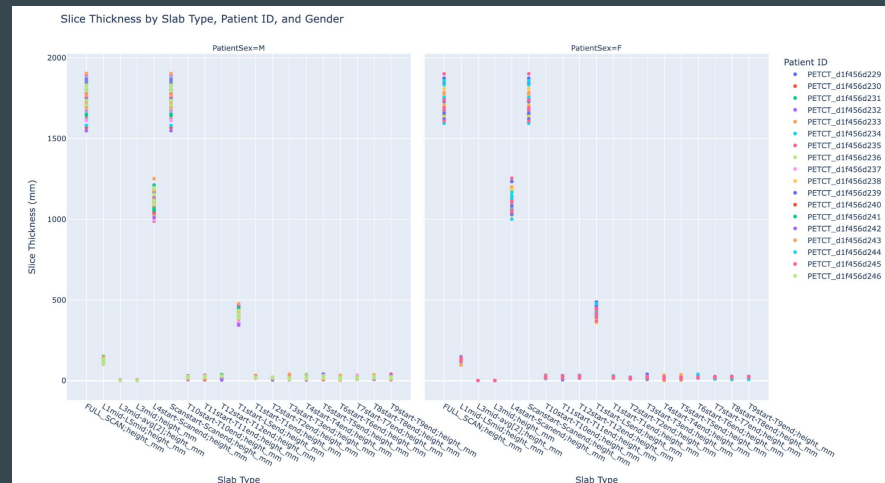
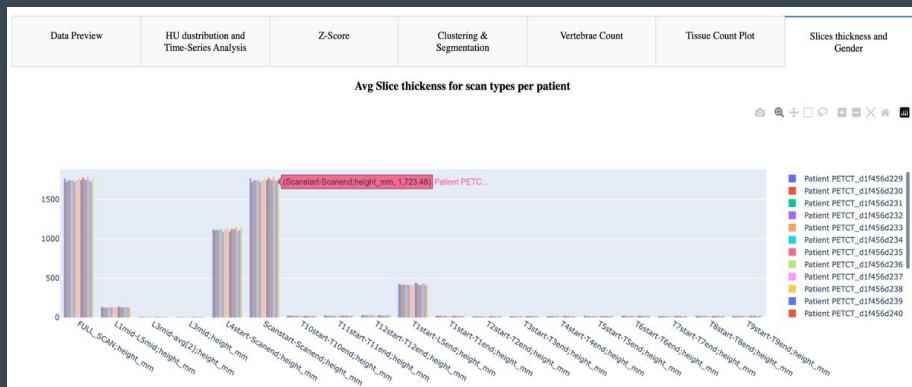
- **Tissue Count plot**

Graphically represent tissue composition, enabling quick insights into different tissue types



- **Slice Thickness and Gender**

Avg. Slice Thickness for patients IDs across scan slabs

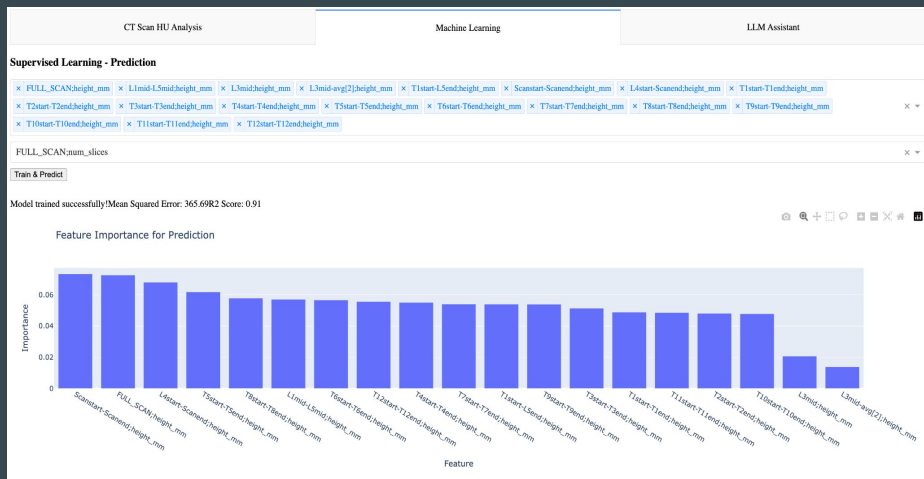


Compare slice thickness metrics segmented by gender for targeted studies.

MACHINE LEARNING PLOTS

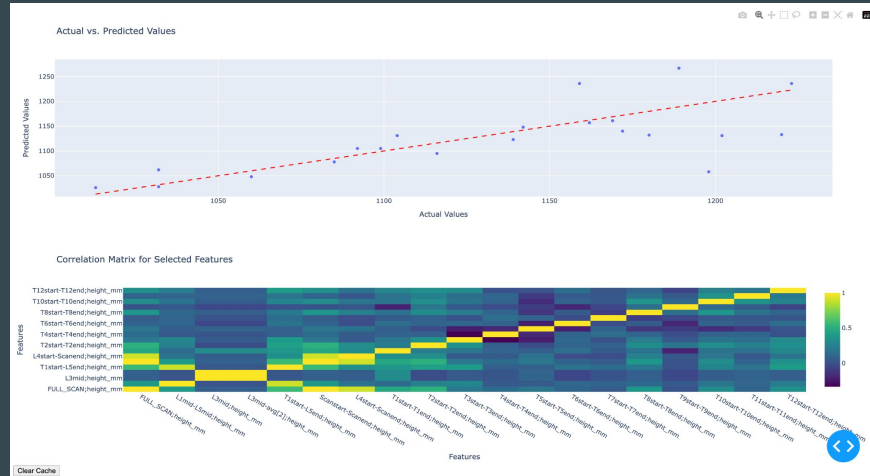
- **Feature Insights**

Explore trends with MSE, R-score, and feature importance graphs



- **Prediction Analysis**

Compare actual vs. predicted values and visualize feature correlations



LLM Assistant

Pre-cached responses added for efficiency, LLM responds to queries within 15- 30 seconds for complex queries

CT Scan HU Analysis	Machine Learning	LLM Assistant
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LLM Assistant

Get Response

Enter a prompt and click 'Get Response'.

Clear Cache

CT Scan HU Analysis	Machine Learning - Clustering	LLM Assistant
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LLM Assistant

Get Response

answer: What are different type of CT scans? Answer: A CT scan is a medical imaging test that uses X-rays to create an image of the inside of your body. It can be used to diagnose diseases such as cancer, heart disease, and stroke. A CT scan involves taking a series of X-ray images of your body's internal organs. These images are then sent to a computer where they are analyzed to determine if there is a problem with your body. Types of CT Scans

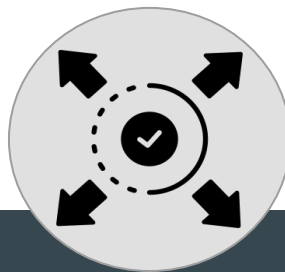
Clear Cache

RESULTS AND OBSERVATIONS



Efficiency

Streamlined Workflow: Automates, segmentation and EDA, reduces manual workload.



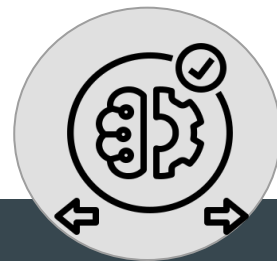
Scalability

Multi-Patient Analysis: Handles large datasets from multiple patients simultaneously.



Accessibility

User-centric: Visualizes complex CT scan into simplified format for non-expert users.



Interpretability

Enhanced Medical Insights: Transforms complex data into clear insights, enabling faster decision-making.

Next steps!

- Scalability: Enable real-time processing for diverse and larger datasets.
- Enhanced Accuracy: Refine clustering & ML prediction robustness
- Deployment: Implement LVM for seamless multi-user access.
- LLM Integration: Expand LLM to directly analyze uploaded data.

Thank you

Questions?